INCH-POUND
MIL-DTL-49285/10A
w/AMENDMENT 1
27 March 2018
SUPERSEDING
MIL DTL 49285/10A
15 May 2009

DETAIL SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL, ONE INDIVIDUALLY SHIELDED PAIR OF 22 (7 X 30) AWG

Inactive for new design after 16 June 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-49285.

Engineering information: See table I.

Note: : Not recommended for shipboard or aerospace applications due to toxic smoke release in the event of a fire. PVC is not recommended in other applications (i.e. confined spaces, low ventilation) where there is the potential for human smoke exposure in a fire.

TABLE I. Description, electrical.

| Electrical characteristics | Requirements |
|---|---|
| Nominal capacitance between conductors | 34 pF/ft (picofarads/foot) |
| Nominal capacitance between the conductor and the shield with one conductor tied to the shield | 67 pF/ft |
| Continuous working voltage | 200 V rms maximum between conductors |
| Shield dc resistance | 14 ohms/1,000 feet maximum |



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TABLE II. Description, physical.

| Components | Construction details |
|---|---|
| Number of pairs | One individually shielded |
| Conductor type and wire size | Stranded tinned copper 22 AWG |
| Conductor stranding | 7 X 30 AWG |
| Conductor insulation | Polypropylene |
| Conductor insulation thickness | 0.008 inch nominal thickness |
| Drain wire type and size | Stranded tinned copper 22 AWG |
| Drain wire stranding | 7 X 30 AWG |
| Jacket material | Polyvinyl chloride |
| Jacket thickness | 0.020 inch |
| Finished cable diameter | 0.142 inch maximum outside diameter |
| Cable style (UL) | Not applicable |
| Tensile strength (jacket) | 2,000 pounds per square inch minimum |
| Elongation (jacket) | 150 percent minimum |
| Overall cabling lay length | 6.8 twists per foot ± 10 percent |

REQUIREMENTS:

Design and construction. See table II.

Shield color code. The shield color shall be blue.

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair, with the aluminum foil side outward. Insulation on the interior of the shield is to be complete with no aluminum available for contact.

Drain wire location. The drain wire is to be spirally located between the shield and jacket, and is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.



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Shield integrity test.*

| Frequency | Response |
|---|---|
| 100 kHz | 109 dB or more below reference |
| 500 kHz | 105 dB or more below reference |
| 1 MHz | 99 dB or more below reference |
| 5 MHz | 85 dB or more below reference |
| 10 MHz | 79 dB or more below reference |
| 15 MHz | 76 dB or more below reference |
| 20 MHz | 72 dB or more below reference |
| 25 MHz | 68 dB or more below reference |
| 30 MHz | 65 dB or more below reference |

* Any two readings may deviate to a value of 0.9 times the stated limits.

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 51 dB below reference.

Flammability. The cable shall pass UL 2556 flame test requirements.

Crosstalk test limits. Crosstalk testing is not required for this configuration.

Durometer hardness. The cable jacket shall have a “Shore A” hardness of 90 $\pm$ 5.

Part or Identifying Number (PIN): The PIN shall be M49285/10.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced document. In addition to MIL-DTL-49285, this document references the following: UL 2556.



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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army – MI
Air Force – 99

(Project 6145-2018-023)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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